

Integrated Features (29–429)

LOPO CV

UNSUPERVISED

Unsupervised Discovery

HVG selection (1,500 genes)

PCA to 50 PCs

SNN graph (n=15) + UMAP

Leiden clusters (n=7)

SUPERVISED

Supervised Prediction

ML: LR, RF, XGBoost, DT

DL: MLP, CNN, BiLSTM, RNN, Transformer

Hyperparameter search

Best: AUC 0.978, Acc 0.928

Interpretation: SHAP + gradient attribution